

Existence of a Partially Revealing Rational-Expectations Equilibrium with Three CRRA Traders and No Noise

REZN Project*

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Abstract

We prove existence of a partially revealing rational-expectations equilibrium (PR-REE) in a fully deterministic asset market with three symmetric CRRA traders, a binary payoff, conditionally Gaussian private signals, and *no* noise traders, liquidity shocks, or random supply. The equilibrium price function is obtained as a Schauder fixed point of the “infer-from-price $\rightarrow$ Bayes-update $\rightarrow$ CRRA-clear” operator on a compact convex set of symmetric, odd-in-logit, uniformly Lipschitz price functions. Three structural lemmas drive the argument: a uniform two-sided bound on the Bayes evidence carried by the price; the fact that CRRA market clearing is a *non-expansive aggregator* of log-posterior-odds, with strict contraction exactly when traders disagree with the price (the Jensen wedge); and the stability of the symmetry class under the operator. The construction works directly with a doubly regularized (price-band, signal-mollified) evidence operator; we show that the regularization breaks the Grossman–Stiglitz degeneracy under which the fully revealing price is always a fixed point of the exact operator, and that the fixed point we construct lies in a subcritical-tilt class that excludes the fully revealing price by an explicit gradient gap. Passage to the exact (co-area) operator is established conditionally on uniform $C^{1,\alpha}$ and gradient bounds that are verified numerically along a certified branch of more than one hundred high-precision equilibria; a continuation corollary in (γ, τ) follows from the implicit function theorem at any certified point.

1 Setup

1.1 Primitives

There are $K = 3$ traders indexed by k , a single risky asset with payoff $v \in \{0, 1\}$, prior $\Pr(v = 1) = \frac{1}{2}$, and a riskless numeraire. Trader k has CRRA utility with common relative risk aversion $\gamma > 0$ and unit wealth $W = 1$. Each trader observes the centred private signal $u_k = s_k - \frac{1}{2}$ where $s_k \mid v \sim N(v, 1/\tau)$ independently across k , $\tau > 0$. Write $u = (u_1, u_2, u_3)$, and

$$f_v(t) = \sqrt{\frac{\tau}{2\pi}} \exp\left(-\frac{\tau}{2}\left(t - \left(v - \frac{1}{2}\right)\right)^2\right), \quad \frac{f_1(t)}{f_0(t)} = e^{\tau t}. \quad (1)$$

Let $\Lambda(x) = (1 + e^{-x})^{-1}$ denote the logistic function and $\text{logit } p = \log \frac{p}{1-p}$ its inverse. There are no noise traders, no supply shocks, and no random endowments: the market-clearing price is a deterministic function of u alone.

*Fixed-Point Factory, REZN archive. Numerical certificates referenced below are in `projects/REZN/solved_fixed_points/` of the project repository.

1.2 CRRA demand and market clearing

A CRRA- γ trader with posterior $\mu \in (0, 1)$ facing price $p \in (0, 1)$ has the closed-form excess demand (in shares, unit wealth)

$$x(\mu, p) = \frac{R - 1}{(1 - p) + Rp}, \quad R = \exp\left(\frac{m - \pi}{\gamma}\right), \quad m = \text{logit } \mu, \quad \pi = \text{logit } p. \quad (2)$$

We freely write $x(m, \pi)$ for (2) in logit coordinates. Given posteriors $(\mu_1, \mu_2, \mu_3) \in (0, 1)^3$, the market-clearing price $\text{clear}(\mu_1, \mu_2, \mu_3) \in (0, 1)$ is the unique root in p of $\sum_{k=1}^3 x(\mu_k, p) = 0$ (uniqueness is part of Lemma 2).

1.3 Price domain and the exact evidence integrals

Fix a truncation level $U > 0$ and set $Q := [-U, U]^3$. For a price function $P : Q \rightarrow (0, 1)$ write $\Pi := \text{logit } P$. When $P \in C^1(Q)$ and the in-slice gradient $\nabla_{(j,l)} P(u_k, \cdot, \cdot)$ does not vanish, the *co-area evidence* of trader k at observed price p is

$$A_v(p, u_k; P) := \int_{\{(t_j, t_l) : P(u_k, t_j, t_l) = p\}} \frac{f_v(t_j) f_v(t_l)}{|\nabla_{(j,l)} P|} d\mathcal{H}^1, \quad (3)$$

where $\mathcal{H}^1$ is one-dimensional Hausdorff measure on the level curve and the weight $1/|\nabla P|$ is the co-area (conditional-density) factor; by the co-area formula [7], $A_v(\cdot, u_k; P)$ is the density of the price $P(u_k, t_j, t_l)$ when $(t_j, t_l) \sim f_v \otimes f_v$, i.e. exactly the likelihood of the price observation under state v . (Omitting the weight yields the arc-length measure, which is *not* the Bayes conditional; in the project archive the unweighted integral is biased by $\approx 19\%$ [9].)

Definition 1 (Exact PR-REE). A measurable $P : Q \rightarrow (0, 1)$ is a *partially revealing rational-expectations equilibrium* if $P \in C^1(Q)$ with non-vanishing in-slice gradients, and for every $u \in Q$ and every k :

(1) the posterior of trader k after observing $(u_k, P = p)$ is

$$\mu_k(u_k, p) = \frac{f_1(u_k) A_1(p, u_k; P)}{f_0(u_k) A_0(p, u_k; P) + f_1(u_k) A_1(p, u_k; P)};$$

(2) markets clear: $\sum_k x(\mu_k(u_k, P(u)), P(u)) = 0$;

(3) consistency: $P(u) = \text{clear}(\mu_1(u_1, P(u)), \mu_2(u_2, P(u)), \mu_3(u_3, P(u)))$;

and P is not fully revealing, i.e. $P \neq P_{\text{FR}}$, where $P_{\text{FR}}(u) := \Lambda(\tau(u_1 + u_2 + u_3))$ is the full-information (sufficient-statistic) price.

1.4 The regularized operator

For arbitrary continuous P the level sets in (3) need not be rectifiable, so we build the fixed-point argument around a doubly regularized evidence functional and recover (3) in the limit (Section 5). Let K_h and ρ_δ be centred Gaussian densities with standard deviations $h > 0$ (logit-price bandwidth) and $\delta > 0$ (own-signal mollification). For bounded measurable $\Pi : Q \rightarrow \mathbb{R}$, $\hat{\pi} \in \mathbb{R}$, $u_1 \in [-U, U]$, define

$$A_v^{h,\delta}(\hat{\pi}, u_1; \Pi) := \int_Q K_h(\Pi(t) - \hat{\pi}) \rho_\delta(t_1 - u_1) f_v(t_2) f_v(t_3) dt, \quad v \in \{0, 1\}, \quad (4)$$

$$b^{h,\delta}(\hat{\pi}, u_1; \Pi) := \log \frac{A_1^{h,\delta}(\hat{\pi}, u_1; \Pi)}{A_0^{h,\delta}(\hat{\pi}, u_1; \Pi)}, \quad \mu_k^{h,\delta}(u_k, p; P) := \Lambda(\tau u_k + b^{h,\delta}(\logit p, u_k; \Pi)). \quad (5)$$

(The own-signal slice $\{t_1 = u_1\}$ is replaced by the mollified weight ρ_δ ; the price level set by the Gaussian band K_h in logit-price. Both are consistent quadratures of (3): as $(h, \delta) \rightarrow 0$ with P smooth and non-degenerate, $A_v^{h,\delta} \rightarrow A_v$; see Lemma 4. The band evidence was verified against the exact co-area integral to 1.5×10^{-5} in the archive [9].)

The equilibrium operator is

$$\Phi_{h,\delta}(P)(u) := \text{clear} \left(\mu_1^{h,\delta}(u_1, P(u); P), \mu_2^{h,\delta}(u_2, P(u); P), \mu_3^{h,\delta}(u_3, P(u); P) \right), \quad u \in Q. \quad (6)$$

A fixed point $P = \Phi_{h,\delta}(P)$ satisfies (1)–(3) of Definition 1 with A_v replaced by $A_v^{h,\delta}$; we call such a P an (h, δ) -regularized REE. We write $\Psi_{h,\delta}(\Pi) := \logit \Phi_{h,\delta}(\Lambda(\Pi))$ for the operator in logit coordinates.

1.5 Symmetry class

Throughout, $\|u\|_\infty = \max_k |u_k|$. Call $\Pi : Q \rightarrow \mathbb{R}$ *symmetric* if invariant under permutations of (u_1, u_2, u_3) and *odd* if $\Pi(-u) = -\Pi(u)$ (equivalently $P(-u) = 1 - P(u)$: the model is invariant under $v \mapsto 1 - v$, $u \mapsto -u$, $p \mapsto 1 - p$). For $B, L > 0$ set

$$\mathcal{K}(B, L) := \left\{ \Pi \in C(Q) : \Pi \text{ symmetric and odd, } \|\Pi\|_\infty \leq B, |\Pi(u) - \Pi(u')| \leq L \|u - u'\|_\infty \right\}. \quad (7)$$

$\mathcal{K}(B, L)$ is nonempty ($\Pi \equiv 0$), convex, and compact in $(C(Q), \|\cdot\|_\infty)$ by Arzelà–Ascoli. Note $\Pi_{\text{FR}} = \tau(u_1 + u_2 + u_3)$ has $\|\cdot\|_\infty$ -Lipschitz constant exactly 3τ (attained along the diagonal): this is the *critical tilt*.

2 Lemmas

Throughout this section fix $\tau, \gamma, U, h, \delta > 0$ and set

$$B := 3\tau U, \quad \chi := \frac{4B}{h^2} = \frac{12\tau U}{h^2}, \quad \beta := \frac{4U}{\delta^2}. \quad (8)$$

Lemma 1 (Bayes representation and evidence bounds). *Let $\Pi : Q \rightarrow \mathbb{R}$ be measurable with $\|\Pi\|_\infty \leq B$ and let $\hat{\pi} \in [-B, B]$, $u_1 \in [-U, U]$. Then:*

(i) (*Representation.*) $\logit \mu_1^{h,\delta} = \tau u_1 + b^{h,\delta}(\hat{\pi}, u_1; \Pi)$, and

$$\frac{A_1^{h,\delta}}{A_0^{h,\delta}}(\hat{\pi}, u_1; \Pi) = \mathbb{E}_\nu [e^{\tau(t_2 + t_3)}], \quad d\nu \propto K_h(\Pi(t) - \hat{\pi}) \rho_\delta(t_1 - u_1) f_0(t_2) f_0(t_3) \mathbf{1}_Q(t) dt.$$

(ii) (*Two-sided evidence bound.*) $|b^{h,\delta}(\hat{\pi}, u_1; \Pi)| \leq 2\tau U$, hence $|\logit \mu_1^{h,\delta}| \leq \tau U + 2\tau U = B$ and $\mu_1^{h,\delta} \in [\varepsilon, 1 - \varepsilon]$ with $\varepsilon := \Lambda(-B) = (1 + e^{3\tau U})^{-1}$.

(iii) (*Smoothness in the conditioning variables.*) $b^{h,\delta}(\cdot, \cdot; \Pi) \in C^\infty([-B, B] \times [-U, U])$ with

$$\left| \partial_{\hat{\pi}} b^{h,\delta} \right| \leq \chi, \quad \left| \partial_{u_1} b^{h,\delta} \right| \leq \beta.$$

(iv) (Stability in Π .) There are $a_0 = a_0(h, \delta, U, \tau) > 0$ and $C_P := 2 \sup |K'_h|/a_0 < \infty$ such that $A_v^{h, \delta} \geq a_0$ and, for any two admissible $\Pi, \tilde{\Pi}$,

$$|b^{h, \delta}(\hat{\pi}, u_1; \Pi) - b^{h, \delta}(\hat{\pi}, u_1; \tilde{\Pi})| \leq C_P \left\| \Pi - \tilde{\Pi} \right\|_{\infty}.$$

Proof. (i) Bayes' rule with prior odds 1 gives posterior odds $\frac{f_1(u_1)A_1}{f_0(u_1)A_0}$; by (1) $\log \frac{f_1(u_1)}{f_0(u_1)} = \tau u_1$. Writing $f_1(t_2)f_1(t_3) = e^{\tau(t_2+t_3)}f_0(t_2)f_0(t_3)$ in (4) yields the stated ratio.

(ii) On Q , $|t_2 + t_3| \leq 2U$, so the expectation in (i) lies in $[e^{-2\tau U}, e^{2\tau U}]$.

(iii) Differentiation under the integral sign is justified by dominated convergence. For the Gaussian kernel $K'_h(z) = -(z/h^2)K_h(z)$, and on the domain of integration $|\Pi(t) - \hat{\pi}| \leq 2B$, hence $|\partial_{\hat{\pi}} A_v| \leq (2B/h^2)A_v$, so $|\partial_{\hat{\pi}} \log A_v| \leq 2B/h^2$ and $|\partial_{\hat{\pi}} b| \leq 4B/h^2 = \chi$. Similarly $\rho'_\delta(z) = -(z/\delta^2)\rho_\delta(z)$ with $|t_1 - u_1| \leq 2U$ gives $|\partial_{u_1} \log A_v| \leq 2U/\delta^2$ and $|\partial_{u_1} b| \leq \beta$.

(iv) Lower bound: $K_h(\Pi(t) - \hat{\pi}) \geq K_h(2B)$ on Q , so $A_v \geq K_h(2B) c_\rho c_f^2 =: a_0$, where $c_\rho := \min_{|u_1| \leq U} \int_{-U}^U \rho_\delta(t - u_1) dt > 0$ and $c_f := \min_v \int_{-U}^U f_v > 0$. Stability: $|A_v(\Pi) - A_v(\tilde{\Pi})| \leq \sup |K'_h| \cdot \left\| \Pi - \tilde{\Pi} \right\|_{\infty}$ (since ρ_δ, f_v integrate to at most one), and $|\log A_v(\Pi) - \log A_v(\tilde{\Pi})| \leq |A_v(\Pi) - A_v(\tilde{\Pi})|/a_0$. $\square$

Lemma 2 (CRRA clearing is a non-expansive aggregator). *Let $x(m, \pi)$ be as in (2) and $D := (1 - p) + Rp > 0$. Then:*

(i) $\partial_m x = \frac{R}{\gamma D^2} > 0$ and $\partial_\pi x = -\partial_m x - \frac{(R-1)^2 p(1-p)}{D^2} \leq -\partial_m x < 0$, with equality iff $R = 1$ (i.e. $m = \pi$); moreover $x(\pi, \pi) = 0$, $x(m, \pi) \rightarrow +\infty$ as $\pi \rightarrow -\infty$ and $x(m, \pi) \rightarrow -\infty$ as $\pi \rightarrow +\infty$, for every fixed m and every $\gamma > 0$.

(ii) For every $(m_1, m_2, m_3) \in \mathbb{R}^3$ the aggregate $Z(\pi) := \sum_k x(m_k, \pi)$ has a unique root $\pi^*(m_1, m_2, m_3)$, and $\min_k m_k \leq \pi^* \leq \max_k m_k$.

(iii) π^* is C^∞ , symmetric, odd ($\pi^*(-m) = -\pi^*(m)$), with

$$\frac{\partial \pi^*}{\partial m_k} = w_k \geq 0, \quad \sum_k w_k = \frac{\sum_k \partial_m x(m_k, \pi^*)}{\sum_k [\partial_m x(m_k, \pi^*) + J_k]} \leq 1, \quad J_k := \frac{(R_k - 1)^2 p(1-p)}{D_k^2},$$

with $\sum_k w_k < 1$ unless $m_1 = m_2 = m_3 = \pi^*$ (the no-trade point). Consequently π^* is non-expansive in the sup norm: $|\pi^*(m) - \pi^*(m')| \leq \|m - m'\|_{\infty}$.

Proof. (i) Write $x = (R-1)/D$ with $R = e^{(m-\pi)/\gamma}$, $p = \Lambda(\pi)$. Then $\partial_R x = [D - (R-1)p]/D^2 = 1/D^2$, so $\partial_m x = (R/\gamma)/D^2 > 0$. For $\partial_\pi x$, note $\partial_\pi R = -R/\gamma$ and $\partial_\pi p = p(1-p)$, while at fixed R , $\partial_p x = -(R-1)^2/D^2$; the chain rule gives the displayed identity. Boundary behaviour: as $\pi \rightarrow +\infty$, $R \rightarrow 0$, $p \rightarrow 1$, $D \rightarrow 0^+$, $x \rightarrow -1/D \rightarrow -\infty$. As $\pi \rightarrow -\infty$, $R \rightarrow \infty$, $p \rightarrow 0$; $Rp \asymp e^{m/\gamma} e^{\pi(1-1/\gamma)}$, so either $Rp \rightarrow 0$ (then $x \approx R-1 \rightarrow +\infty$) or $Rp \rightarrow \infty$ (then $x \approx 1/p \rightarrow +\infty$) or $Rp \rightarrow e^m$ ($\gamma = 1$; then $x \rightarrow +\infty$ since $R \rightarrow \infty$). Finally $m = \pi$ gives $R = 1$, $x = 0$.

(ii) Z is continuous, strictly decreasing by (i), and ranges over all of $\mathbb{R}$, so the root exists and is unique. At $\pi = \max_k m_k$ each $x(m_k, \pi) \leq 0$ by (i) and $x(\pi, \pi) = 0$, so $Z \leq 0$ and $\pi^* \leq \max_k m_k$; symmetrically $\pi^* \geq \min_k m_k$.

(iii) Smoothness is the implicit function theorem ($\partial_\pi Z < 0$ strictly). Implicit differentiation gives $w_k = \partial_m x(m_k, \pi^*) / \sum_j (-\partial_\pi x(m_j, \pi^*))$, and $-\partial_\pi x = \partial_m x + J$ from (i) yields the displayed sum, which is ≤ 1 with equality iff every $J_k = 0$, i.e. every $R_k = 1$. Oddness: with $(m, \pi) \mapsto (-m, -\pi)$ one has $R \mapsto 1/R$, $p \mapsto 1-p$, and a direct computation gives $x(-m, -\pi) = -x(m, \pi)$, so Z flips sign and the root is odd. Non-expansiveness follows from the mean value theorem: $|\pi^*(m) - \pi^*(m')| \leq (\sum_k w_k) \|m - m'\|_\infty \leq \|m - m'\|_\infty$. $\square$

Remark 1 (Jensen wedge). The quantity J_k in Lemma 2 is the curvature (wealth-effect) term absent under CARA; a second-order expansion of $Z(\pi) = 0$ gives the project's closed form $\pi^* \approx \bar{m} + (\frac{1}{2} - p) \text{Var}(m)/\gamma$: the price is the average log-odds plus a wedge proportional to belief *disagreement* and inversely proportional to γ . The wedge vanishes identically under full revelation ($m_k \equiv \pi$), which is the analytic root of the Grossman–Stiglitz degeneracy in Proposition 1 below [9].

Lemma 3 (Invariance and compactness of the operator). *Assume $\chi < 1$ and set $L_\star := \frac{\tau + \beta}{1 - \chi}$, $\mathcal{K} := \mathcal{K}(B, L_\star)$ as in (7). Then $\Psi_{h,\delta}$ (the operator (6) in logit coordinates) maps $\mathcal{K}$ into $\mathcal{K}$ and is Lipschitz continuous on $(\mathcal{K}, \|\cdot\|_\infty)$ with constant $\chi + C_P$.*

Proof. Fix $\Pi \in \mathcal{K}$ and write

$$m_k(u) := \tau u_k + b^{h,\delta}(\Pi(u), u_k; \Pi), \quad \Psi(\Pi)(u) = \pi^*(m_1(u), m_2(u), m_3(u)).$$

Range. By Lemma 1(ii), $|m_k(u)| \leq B$, and by the sandwich in Lemma 2(ii), $|\Psi(\Pi)(u)| \leq B$.

Symmetry. If σ permutes coordinates, then since Π is symmetric, $\Pi(u^\sigma) = \Pi(u)$ and $m_k(u^\sigma) = m_{\sigma(k)}(u)$; π^* is symmetric, so $\Psi(\Pi)(u^\sigma) = \Psi(\Pi)(u)$.

Oddness. Substituting $t \mapsto -t$ in (4) and using K_h even, Π odd, ρ_δ even, $f_v(-t) = f_{1-v}(t)$, one gets $A_v^{h,\delta}(-\hat{\pi}, -u_1; \Pi) = A_{1-v}^{h,\delta}(\hat{\pi}, u_1; \Pi)$, hence $b(-\hat{\pi}, -u_1; \Pi) = -b(\hat{\pi}, u_1; \Pi)$ and $m_k(-u) = -m_k(u)$. Oddness of π^* (Lemma 2(iii)) finishes.

Lipschitz bound. For $u, u' \in Q$, by Lemma 1(iii) and $\text{Lip}_\infty(\Pi) \leq L_\star$,

$$|m_k(u) - m_k(u')| \leq \tau |u_k - u'_k| + \chi |\Pi(u) - \Pi(u')| + \beta |u_k - u'_k| \leq (\tau + \beta + \chi L_\star) \|u - u'\|_\infty.$$

By non-expansiveness (Lemma 2(iii)), $|\Psi(\Pi)(u) - \Psi(\Pi)(u')| \leq \max_k |m_k(u) - m_k(u')|$, and $\tau + \beta + \chi L_\star = L_\star$ by the definition of $L_\star$. So $\Psi(\Pi) \in \mathcal{K}$.

Continuity in Π . For $\Pi, \tilde{\Pi} \in \mathcal{K}$ and any u ,

$$\begin{aligned} |m_k(u) - \tilde{m}_k(u)| &\leq |b(\Pi(u), u_k; \Pi) - b(\tilde{\Pi}(u), u_k; \Pi)| + |b(\tilde{\Pi}(u), u_k; \Pi) - b(\tilde{\Pi}(u), u_k; \tilde{\Pi})| \\ &\leq (\chi + C_P) \|\Pi - \tilde{\Pi}\|_\infty \end{aligned}$$

by Lemma 1(iii)–(iv), and non-expansiveness of π^* transfers the bound to Ψ . $\square$

3 Main existence theorem

Theorem 1 (Existence of a regularized PR-REE). *Fix $\tau, \gamma, U > 0$ and regularization parameters $h, \delta > 0$ with*

$$\chi = \frac{12\tau U}{h^2} < 1.$$

Let $B = 3\tau U$, $\beta = 4U/\delta^2$, $L_\star = (\tau + \beta)/(1 - \chi)$, and $\mathcal{K} = \mathcal{K}(B, L_\star)$ as in (7). Then the operator $\Phi_{h,\delta}$ of (6) has a fixed point $P_\star = \Lambda(\Pi_\star)$ with $\Pi_\star \in \mathcal{K}$. The price function $P_\star$ takes values in $[\varepsilon, 1 - \varepsilon]$ with $\varepsilon = (1 + e^{3\tau U})^{-1}$, is symmetric in (u_1, u_2, u_3) , satisfies $P_\star(-u) = 1 - P_\star(u)$, and satisfies conditions (1)–(3) of Definition 1 with the evidence A_v replaced by $A_v^{h,\delta}$, i.e. it is an (h, δ) -regularized REE. If in addition

$$\chi \leq \frac{1}{3} \quad \text{and} \quad \beta < \tau(2 - 3\chi), \quad (9)$$

then $L_\star < 3\tau = \text{Lip}_\infty(\Pi_{\text{FR}})$, so the entire candidate set $\mathcal{K}$ — and in particular $P_\star$ — lies in the subcritical-tilt class that excludes the fully revealing price: $P_\star \neq P_{\text{FR}}$.

Proof. $\mathcal{K}$ is a nonempty, convex subset of the Banach space $(C(Q), \|\cdot\|_\infty)$; it is bounded and uniformly equicontinuous (Lipschitz constant $L_\star$), and closed under uniform limits, hence compact by Arzelà–Ascoli. By Lemma 3, $\Psi_{h,\delta} : \mathcal{K} \rightarrow \mathcal{K}$ is well defined and Lipschitz continuous. Schauder’s fixed point theorem [8, Cor. 2.13] yields $\Pi_\star \in \mathcal{K}$ with $\Psi_{h,\delta}(\Pi_\star) = \Pi_\star$, i.e. $P_\star := \Lambda(\Pi_\star)$ satisfies $\Phi_{h,\delta}(P_\star) = P_\star$. Unwinding (6): at every $u \in Q$, condition (3) holds with $p = P_\star(u)$, and since clear outputs the unique market-clearing root, (2) holds; (1) holds with the regularized evidence by construction (5). The range, symmetry and oddness statements are $\Pi_\star \in \mathcal{K}$ plus Lemma 1(ii). Finally, under (9), $L_\star = (\tau + \beta)/(1 - \chi) < \tau(3 - 3\chi)/(1 - \chi) = 3\tau$, while $\Pi_{\text{FR}} = \tau(u_1 + u_2 + u_3)$ satisfies $|\Pi_{\text{FR}}(u) - \Pi_{\text{FR}}(u')| = 3\tau \|u - u'\|_\infty$ along the diagonal direction, so $\Pi_{\text{FR}} \notin \mathcal{K}$ and $\Pi_\star \neq \Pi_{\text{FR}}$. $\square$

Remark 2 (Reading the hypotheses). The single substantive hypothesis is the *subcritical feedback* condition $\chi < 1$: the sensitivity of the price-evidence to the conditioning price must not exceed unity, otherwise the inference channel can amplify price variation without bound (this is precisely the self-referential channel that produces the fully revealing solution). The condition is a sufficient worst-case bound, not sharp: it requires $h^2 > 12\tau U$, i.e. a coarse price-band relative to the logit range $[-3\tau U, 3\tau U]$. Sharper (measure-weighted, rather than sup-norm) estimates of $\partial_{\hat{\pi}} b$ would relax it; the numerical continuation of Corollary 1 is how the result is transported to small h in practice.

4 The Grossman–Stiglitz degeneracy and its resolution

At exact $h = \delta = 0$ the fully revealing price is *always* a fixed point; this is the noiseless-REE degeneracy of [2, 3, 1]. The regularization breaks it. Define the exact ($h = \delta = 0$) operator Φ_0 by (6) with the co-area evidence (3) (well defined at P_{FR} , whose in-slice gradients never vanish).

Proposition 1 (Degeneracy at $h = 0$; broken for $h, \delta > 0$). (i) $\Phi_0(P_{\text{FR}}) = P_{\text{FR}}$ on Q : the fully revealing price is an exact fixed point of the unregularized operator for every (γ, τ, U) .

(ii) For every $h, \delta > 0$ with $h^2 + \tau^2 \delta^2 > 0$ there exists $U_0 = U_0(h, \delta, \tau, \gamma)$ such that for all $U \geq U_0$, $\Phi_{h,\delta}(P_{\text{FR}}) \neq P_{\text{FR}}$. In particular no fixed point of $\Phi_{h,\delta}$ equals P_{FR} .

Proof. (i) Fix u_1 and $\hat{\pi} = \text{logit } p$. On the level set $\{(t_2, t_3) : \Pi_{\text{FR}}(u_1, t_2, t_3) = \hat{\pi}\}$ the others’ aggregate $\tau(t_2 + t_3) = \hat{\pi} - \tau u_1$ is *constant*. Hence in the ratio A_1/A_0 , whatever measure is used on the level set (the co-area weight cancels), $b(\hat{\pi}, u_1) = \log e^{\hat{\pi} - \tau u_1} = \hat{\pi} - \tau u_1$ exactly, so $m_k = \tau u_k + b = \hat{\pi}$ for every k : each trader’s posterior equals the observed price, every excess

demand is $x(\hat{\pi}, \hat{\pi}) = 0$ (Lemma 2(i)), and the market clears at the observed price itself. Thus $\Phi_0(P_{\text{FR}})(u) = P_{\text{FR}}(u)$ for all u . (The archive verifies this numerically to residual 1.7×10^{-15} [9].)

(ii) First take $U = \infty$ in (4) (all integrals over $\mathbb{R}^3$; absolutely convergent). At $\Pi = \Pi_{\text{FR}}$ everything is Gaussian: under ν of Lemma 1(i), $w := \tau(t_2 + t_3) \sim N(-\tau, 2\tau)$ a priori (under $f_0 \otimes f_0$), $\tau t_1 \sim N(\tau u_1, \tau^2 \delta^2)$, and the kernel conditions on the noisy observation $\tau t_1 + w = \hat{\pi} + N(0, h^2)$. Gaussian conditioning plus $\log \mathbb{E} e^w = \mathbb{E} w + \frac{1}{2} \text{Var}(w)$ give, after simplification,

$$m_k = \tau u_k + b^{h,\delta}(\hat{\pi}, u_k; \Pi_{\text{FR}}) = \rho \hat{\pi} + (1 - \rho) \tau u_k, \quad \rho := \frac{2\tau}{2\tau + \tau^2 \delta^2 + h^2} \in (0, 1). \quad (10)$$

So the regularized price-signal is *shrunk*: posteriors are a strict convex combination of the price and the private signal. Evaluate at the test point $u^\dagger = (a, a, -2a)$, $a > 0$, where $\hat{\pi} = \Pi_{\text{FR}}(u^\dagger) = 0$, $p = \frac{1}{2}$. With $c := (1 - \rho)\tau a > 0$ the posterior log-odds are $(c, c, -2c)$ and, using $x(m, 0) = 2 \tanh(m/2\gamma)$, aggregate excess demand at the observed price is

$$Z(0) = 4 \tanh \theta - 2 \tanh 2\theta = \frac{4 \tanh^3 \theta}{1 + \tanh^2 \theta} > 0, \quad \theta := \frac{c}{2\gamma} > 0.$$

Since Z is strictly decreasing (Lemma 2), the clearing root is strictly positive: $\Phi_{h,\delta}(P_{\text{FR}})(u^\dagger) > \frac{1}{2} = P_{\text{FR}}(u^\dagger)$. For finite U the evidence integrals differ from their $U = \infty$ values by Gaussian tail terms: the integrands are dominated by $K_h \rho_\delta f_v f_v \leq \sup K_h \cdot \rho_\delta f_v f_v$, whose mass outside Q vanishes as $U \rightarrow \infty$, while the denominators stay bounded below for $\hat{\pi}$ in a fixed compact set; hence $b_{(U)}^{h,\delta} \rightarrow b_{(\infty)}^{h,\delta}$ uniformly on compacts as $U \rightarrow \infty$. Choosing a fixed (say $a = 1$) and U_0 large enough that the perturbation of $Z(0)$ is less than $\frac{1}{2} \cdot 4 \tanh^3 \theta / (1 + \tanh^2 \theta)$ preserves the strict sign. $\square$

Remark 3 (Risk aversion selects the branch). The displaced demand in the proof of (ii) is $O(\theta^3)$ with $\theta \propto (1 - \rho)\tau a / \gamma$: the force that pushes the price off the fully revealing ray is the *disagreement-times-curvature* (Jensen) channel of Remark 1, scaled by $1/\gamma$. In the CARA limit (demand linear in log-odds) the analogous computation gives $Z(0) = \sum_k (m_k - \hat{\pi}) / \gamma_{\text{CARA}} = 0$ at every point: regularized or not, CARA clearing reproduces the sufficient statistic and the equilibrium is fully revealing — consistent with Hellwig’s closed form, verified in the archive (deficit at P_{FR} of order 10^{-4} and shrinking with resolution; CRRA deficit $\sim \gamma^{-0.94} \rightarrow 0$ as $\gamma \rightarrow \infty$) [3, 9]. Partial revelation here is a CRRA wealth-curvature effect, not a noise effect.

5 The exact co-area limit

We now relate the regularized fixed points to Definition 1. The first lemma is the consistency of the band-mollifier quadrature.

Lemma 4 (Co-area consistency). *Let $\Pi \in C^1(Q)$ be symmetric with $\partial_2 \Pi \geq c > 0$ and $\partial_3 \Pi \geq c > 0$ on Q . Fix $u_1 \in (-U, U)$ and $\hat{\pi}$ an interior value of $\Pi(u_1, \cdot, \cdot)$. Then*

$$A_v^{h,\delta}(\hat{\pi}, u_1; \Pi) \longrightarrow g_v(\hat{\pi}; u_1) := \int_{\{(t_2, t_3): \Pi(u_1, t_2, t_3) = \hat{\pi}\}} \frac{f_v(t_2) f_v(t_3)}{|\nabla_{(2,3)} \Pi|} d\mathcal{H}^1 \quad \text{as } (h, \delta) \rightarrow (0, 0),$$

the co-area evidence computed on the logit surface. Since level sets of Π and of $P = \Lambda(\Pi)$ coincide and the smooth change of level variable rescales g_0 and g_1 by the same positive factor

$\Lambda'(\hat{\pi})^{-1}$, the ratio A_1/A_0 — hence the posterior (5) — converges to its value under the exact evidence (3).

Proof sketch. Let $g_v(\pi; t_1) := \int_{\{\Pi(t_1, \cdot, \cdot) = \pi\}} f_v f_v / |\nabla_{(2,3)} \Pi| d\mathcal{H}^1$ denote the in-slice co-area density. By the co-area formula [7, §3.4],

$$A_v^{h,\delta}(\hat{\pi}, u_1; \Pi) = \int \rho_\delta(t_1 - u_1) \left[\int K_h(\pi - \hat{\pi}) g_v(\pi; t_1) d\pi \right] dt_1.$$

Under the gradient hypothesis the level set $\{\Pi(t_1, \cdot, \cdot) = \pi\}$ is the graph $t_3 = \eta(t_2; \pi, t_1)$ of a C^1 function with slope $-\partial_2 \Pi / \partial_3 \Pi$ bounded by c and $\|\nabla \Pi\|_\infty / c$; its t_2 -domain is an interval whose endpoints solve $\Pi(t_1, t_2, \pm U) = \pi$ and are Lipschitz in (π, t_1) by the implicit function theorem (monotonicity in t_2 from $\partial_2 \Pi \geq c$). Hence $g_v(\cdot; \cdot)$ is continuous, and $K_h * g_v \rightarrow g_v$, $\rho_\delta * \cdot \rightarrow$ evaluation at u_1 , uniformly on compacts. The change of variables between level sets of Π and of $P = \Lambda(\Pi)$ multiplies g_v by the same positive factor for $v = 0, 1$, so the posterior (5) is unaffected. $\square$

To pass to the limit along a family of regularized equilibria we need uniform regularity, which Theorem 1 does not supply as $h \rightarrow 0$ (its hypothesis $\chi < 1$ fails). We therefore state the limit theorem conditionally; the hypotheses are exactly the bounds observed, uniformly, along the certified numerical branch [9].

Assumption 1. There are sequences $h_n, \delta_n \downarrow 0$ and fixed points $P_n = \Lambda(\Pi_n)$ of Φ_{h_n, δ_n} , and constants $\alpha \in (0, 1]$, $C_1 < \infty$, $c > 0$, $\kappa \in (0, 1)$, such that: (A1) $\|\Pi_n\|_{C^{1,\alpha}(Q)} \leq C_1$ and $\partial_k \Pi_n \geq c$ on Q for all k, n ; (A2) (subcritical tilt) $\max_k \|\partial_k \Pi_n\|_\infty \leq \kappa \tau$ for all n .

Theorem 2 (Exact PR-REE as a regularization limit). *Under Assumption 1 there is a subsequence $\Pi_{n_j} \rightarrow \Pi_*$ in $C^1(Q)$ such that $P_* = \Lambda(\Pi_*)$ satisfies conditions (1)–(3) of Definition 1 with the exact co-area evidence (3) at every interior u . Moreover $\max_k \|\partial_k \Pi_*\|_\infty \leq \kappa \tau < \tau$, whereas $\partial_k \Pi_{\text{FR}} \equiv \tau$; hence $P_* \neq P_{\text{FR}}$ and P_* is an exact PR-REE on Q .*

Proof. By (A1) and Arzelà–Ascoli in C^1 (compact embedding $C^{1,\alpha} \hookrightarrow C^1$), extract $\Pi_{n_j} \rightarrow \Pi_*$ in $C^1(Q)$; the bounds $\partial_k \Pi_* \geq c$ and $\|\partial_k \Pi_*\|_\infty \leq \kappa \tau$ pass to the limit. The in-slice co-area densities $g_v^{(n)}$ of Π_n admit the uniform graph parameterization of Lemma 4 with constants depending only on (c, C_1) ; hence $(g_v^{(n)})_n$ is uniformly bounded and equicontinuous, and $g_v^{(n)} \rightarrow g_v^{(*)}$ uniformly on compact subsets of interior level values. Consequently

$$A_v^{h_n, \delta_n}(\hat{\pi}, u_1; \Pi_n) = (\rho_{\delta_n} * (K_{h_n} * g_v^{(n)}))(\hat{\pi}, u_1) \longrightarrow g_v^{(*)}(\hat{\pi}, u_1)$$

uniformly on compacts, with $g_v^{(*)}$ the exact co-area evidence of Π_* . The posteriors $\mu_k^{(n)}$ converge uniformly to the exact co-area posteriors of P_* (the denominators are bounded below by the uniform two-sided evidence bound, Lemma 1(ii), which is h, δ -independent), and clear is continuous (Lemma 2); passing to the limit in $P_n(u) = \Phi_{h_n, \delta_n}(P_n)(u)$ gives (1)–(3) for P_* . The tilt gap gives $\Pi_* \neq \Pi_{\text{FR}}$. $\square$

6 Continuation in $(h, \delta, \gamma, \tau)$

Corollary 1 (Existence by continuation). *Let $z = (h, \delta, \gamma, \tau)$ range in an open set on which χ may exceed 1. The map $(P, z) \mapsto \Phi_z(P)$ is C^1 from $C^{0,1}(Q) \times \mathbb{R}_{++}^4$ to $C^{0,1}(Q)$ (differentiation*

under the integral in (4), smoothness of clear by Lemma 2(iii)). If $\bar{P}$ is a fixed point of $\Phi_{\bar{z}}$ at which $I - D_P\Phi_{\bar{z}}(\bar{P})$ is boundedly invertible, then there is a neighborhood of $\bar{z}$ and a unique C^1 branch $z \mapsto P(z)$ of fixed points through $\bar{P}$. In particular, existence at any certified parameter point extends to an open neighborhood in (γ, τ) and along the regularization path $h \downarrow 0$ for as long as the linearization remains invertible.

Proof. Implicit function theorem in Banach space applied to $F(P, z) := P - \Phi_z(P)$. $\square$

Remark 4 (Numerical instantiation). The archive provides the anchors: the low- τ ladder alone certifies 66 equilibria on $\tau \in \{0.05, 0.1, 0.3, 0.4\} \times \gamma \in [0.05, 30]$ with double-precision residuals $\|P - \Phi(P)\| \leq 8 \times 10^{-15}$ and long-double residuals $\leq 10^{-17}$ (`lowtau/lowtau.json`), and the full (γ, τ) archive contains well over one hundred certified fixed points, several to 100-decimal precision; the joint limit $h \rightarrow 0$, $\Delta x \rightarrow 0$ nails the equilibrium to residual $\sim 10^{-14}$ with a positive, grid- and bandwidth-invariant information deficit $1 - R^2 \approx 0.26$ – 0.28 at $(\gamma, \tau) = (0.1, 2)$ [9]. Quadratic Newton convergence at each anchor is numerical evidence that $I - D_P\Phi$ is invertible there, i.e. that the branch of Corollary 1 passes through every certified point; the positive deficit separates the branch from P_{FR} (whose deficit is 0) uniformly along the sweep. A computer-assisted (interval-arithmetic) bound on $\|(I - D_P\Phi)^{-1}\|$ at one anchor would upgrade Assumption 1 and the branch separation to a fully rigorous theorem at small h ; this is the natural next step.

Remark 5 (Comparative statics). Along any C^1 branch from Corollary 1, the deficit $1 - R^2$ and the price surface vary smoothly in (γ, τ) . Differentiating the second-order clearing identity of Remark 1 predicts the deficit to be decreasing in γ and increasing in τ (the wedge is $(\frac{1}{2} - p)\text{Var}(m)/\gamma$, and $\text{Var}(m)$ grows with signal precision); this is exactly the monotonicity observed on the certified (γ, τ) map [9]. We state it as a numerical regularity rather than a theorem, since the sign argument is second-order exact only in the small-disagreement regime.

7 Discussion

What is proven. Theorem 1 is unconditional: for every (γ, τ, U) and every regularization with subcritical feedback $\chi < 1$, a symmetric, odd, uniformly Lipschitz regularized REE exists, with prices in $[\varepsilon, 1 - \varepsilon]$, and under the slightly stronger condition (9) it lives in a tilt class that provably excludes the fully revealing price. Proposition 1 shows the exact operator always admits P_{FR} (the Grossman–Stiglitz degeneracy, verified in the archive to 1.7×10^{-15}) and that any positive regularization destroys this fixed point, so the object of Theorem 1 is never the revealing price in disguise. Theorem 2 transfers existence to the exact co-area operator of Definition 1, conditionally on uniform $C^{1,\alpha}$, gradient-lower, and subcritical-tilt bounds along a vanishing-regularization sequence — precisely the bounds measured, stably, on the certified numerical branch.

The role of the truncation $[\varepsilon, 1 - \varepsilon]$ and $[-U, U]$. The price floor is not an assumption but an output: the two-sided evidence bound (Lemma 1(ii)) caps the posterior log-odds at $3\tau U$ because three bounded signals carry bounded likelihood ratio. On unbounded signal space the equilibrium price must approach 0 and 1 in the tails, so no uniform floor can hold on $\mathbb{R}^3$; the natural route to an untruncated statement is an exhaustion $U_n \uparrow \infty$ with locally uniform

versions of (A1)–(A2) and a diagonal argument. We leave this as an open (technical, not conceptual) extension.

Why risk aversion matters. The selection of a non-revealing branch is driven entirely by the CRRA wealth-curvature term J_k in Lemma 2: clearing contracts log-odds strictly ($\sum_k w_k < 1$) exactly when traders disagree with the price, and the second-order wedge $(\frac{1}{2} - p)\text{Var}(m)/\gamma$ compresses the price toward $\frac{1}{2}$, i.e. destroys the sufficient statistic. Under CARA the wedge is identically zero, clearing is an exact average of log-odds, and full revelation is the unique outcome (Hellwig’s closed form, reproduced numerically in the archive); accordingly the CRRA deficit vanishes like $\gamma^{-0.94}$ in the CARA limit. The economics: with wealth effects, prices aggregate beliefs *non-linearly*, and a non-degenerate amount of private information remains private — no noise traders are needed to break the Grossman–Stiglitz paradox.

Why the smooth function space is essential. The co-area evidence is well defined only on level sets of positive regularity, and the archive’s diagnosis is sharper: the exact evidence $A_v(\cdot, u_1)$ is C^0 but generically *not* C^1 at Morse-critical prices (where ∇P vanishes on a level set), with derivative blow-ups of two orders of magnitude on controlled test surfaces. The gradient lower bound in (A1) excludes critical levels on Q and is what makes Lemma 4 and Theorem 2 run; it is also why the kernel band should be read as a consistent quadrature of the co-area integral (a numerical-regularity device), not as economic noise: the equilibrium deficit is invariant to h once resolved [9].

Toward uniqueness. Global uniqueness is false at $h = 0$ (P_{FR} always coexists). The right target is uniqueness *within the subcritical-tilt class*. Two routes appear viable: (a) a quantitative version of Lemma 2(iii) — clearing is a strict ℓ^∞ contraction off the no-trade set, with modulus $1 - \min_k J_k/(\cdot)$ — combined with a bound $\chi < 1 - \sup \sum_k w_k$ -type estimate would make Ψ a contraction on $\mathcal{K}$ and give uniqueness and constructive (Picard) convergence; (b) a degree-theoretic argument: if $I - D_P\Phi$ is invertible along the whole branch (as the Newton certificates suggest), the Leray–Schauder degree of $I - \Phi$ on the subcritical class is constant in (h, γ, τ) and equal to ± 1 , ruling out branch folds but not disconnected solutions. Either upgrade requires spectral information about $D_P\Phi$ that is currently numerical.

References

- [1] R. Radner, Rational expectations equilibrium: generic existence and the information revealed by prices, *Econometrica* **47** (1979), 655–678.
- [2] S. J. Grossman and J. E. Stiglitz, On the impossibility of informationally efficient markets, *American Economic Review* **70** (1980), 393–408.
- [3] M. F. Hellwig, On the aggregation of information in competitive markets, *Journal of Economic Theory* **22** (1980), 477–498.
- [4] B. Allen, Generic existence of completely revealing equilibria for economies with uncertainty when prices convey information, *Econometrica* **49** (1981), 1173–1199.

- [5] P. Milgrom and N. Stokey, Information, trade and common knowledge, *Journal of Economic Theory* **26** (1982), 17–27.
- [6] B. Breon-Drish, On existence and uniqueness of equilibrium in a class of noisy rational expectations models, *Review of Economic Studies* **82** (2015), 868–921.
- [7] L. C. Evans and R. F. Gariepy, *Measure Theory and Fine Properties of Functions*, CRC Press, 1992.
- [8] E. Zeidler, *Nonlinear Functional Analysis and its Applications I: Fixed-Point Theorems*, Springer, 1986.
- [9] REZN Project, *K=3 CRRA rational-expectations fixed-point archive*: FINDINGS.md, k3_coarea_limit, k3_verify_coarea, k3_cara/HELLWIG_CLOSED_FORM.md, k3_coarea_2dsweep, lowtau/lowtau.json, Fixed-Point Factory repository, 2026.